

KEOKUK, IA, USA

WMO: 725456

Lat: 40.460N

Lon: 91.428W

Elev: 671

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.4	-8.4	3.6	3.1	-4.1	4.5	8.1	25.6	27.7	23.3	29.7	6.7	320	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	93.3	77.6	90.7	76.8	88.4	75.5	80.4	90.0	78.7	88.3	77.3	86.1	8.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	146.3	86.3	75.5	136.9	84.8	73.6	128.4	83.2	44.5	88.8	42.9	88.2	41.1	86.5	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.9	16.9	-7.7	96.9	6.7	3.2	-12.5	99.2	-16.4	101.1	-20.1	102.9	-25.0	105.3		
(5)			-7.8	82.6	6.4	1.8	-12.4	83.8	-16.1	84.9	-19.7	85.9	-24.4	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	27.2	30.7	42.4	53.9	64.4	73.1	77.1	75.3	67.9	55.7	43.1	31.9		
	DBStd	19.81	12.20	11.75	11.64	9.70	8.25	5.90	5.86	5.40	8.10	9.50	10.48	10.53		
	HDD50	2496	710	548	294	68	3	0	0	0	1	51	256	565		
	HDD65	5375	1173	961	705	352	116	7	1	2	60	316	657	1025		
	CDD50	3836	2	7	58	183	448	694	841	784	539	226	50	4		
	CDD65	1243	0	0	4	18	96	251	377	321	148	26	2	0		
	CDH74	11623	0	0	26	183	828	2297	3926	2860	1289	209	5	0		
	CDH80	4264	0	0	1	30	217	826	1660	1062	425	43	0	0		
(14) Wind	WSAvg	7.5	8.5	8.6	9.2	9.4	7.7	6.4	5.6	5.4	5.9	7.4	8.3	8.1		
(15) Precipitation	PrecAvg	36.9	1.4	1.1	2.8	3.6	4.5	3.9	4.6	3.2	4.5	3.2	2.5	1.8		
	PrecMax	56.1	8.6	3.0	7.4	8.0	10.4	11.2	12.2	10.0	12.7	8.3	8.2	5.7		
	PrecMin	18.2	0.0	0.1	0.4	0.7	0.6	0.2	0.6	1.3	0.0	0.1	0.3	0.1		
	PrecStd	9.0	1.6	0.6	1.6	1.9	2.7	2.4	3.0	2.0	3.1	2.3	1.7	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.4	66.4	78.5	83.8	90.1	92.9	99.3	97.0	93.0	85.6	73.3	64.0		
		MCWB	53.5	54.8	63.8	66.5	74.6	76.6	77.0	74.4	75.2	72.6	58.8	57.4		
	2%	DB	54.3	59.0	72.0	79.1	85.2	90.4	94.7	92.6	89.5	79.4	68.4	55.3		
		MCWB	47.9	50.7	60.7	62.9	72.1	76.1	79.2	77.3	74.2	65.3	58.3	49.2		
	5%	DB	48.3	52.4	65.9	74.8	81.7	88.2	91.1	89.9	85.5	75.1	63.7	50.9		
		MCWB	43.1	46.5	56.3	61.2	68.6	75.2	77.9	76.8	73.7	63.8	55.2	45.8		
	10%	DB	43.3	46.3	60.8	70.4	79.0	85.0	88.4	86.0	81.4	71.8	59.2	46.0		
		MCWB	38.2	41.3	51.8	58.7	66.9	73.5	76.3	75.1	69.8	62.3	50.8	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	58.4	66.5	69.0	77.1	80.4	83.5	82.7	78.3	73.2	63.3	59.5		
		MCDB	61.0	64.0	75.6	79.5	86.9	89.2	93.4	92.1	88.1	82.5	69.6	64.4		
	2%	WB	48.7	51.4	61.8	65.7	73.7	78.2	81.0	80.0	76.3	69.6	59.8	51.3		
		MCDB	53.3	57.8	68.7	76.1	82.3	87.5	90.3	89.5	85.6	76.0	66.2	54.6		
	5%	WB	42.8	46.3	57.7	63.0	71.2	76.6	79.3	78.2	74.2	66.4	56.1	46.2		
		MCDB	48.3	51.9	65.3	72.3	80.1	85.6	89.6	86.9	82.1	72.8	63.5	49.6		
	10%	WB	38.8	41.5	52.3	59.9	68.6	74.7	77.7	76.5	72.1	62.9	51.7	41.9		
		MCDB	43.3	46.7	59.4	68.8	76.4	83.2	86.8	84.0	80.0	69.9	58.6	46.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	15.9	18.0	19.7	18.7	18.4	17.9	17.9	19.9	18.9	16.9	14.3		
		MCDBR	22.5	23.1	24.6	25.2	21.2	20.5	19.9	21.3	21.2	22.9	22.6	19.9		
	5% WB	MCWBR	18.0	16.8	15.4	14.6	12.1	11.1	10.0	11.0	10.0	12.9	15.5	16.0		
		MCWBR	21.0	21.6	21.9	21.9	19.2	18.6	18.2	17.9	18.4	18.6	21.0	18.6		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.310	0.339	0.392	0.425	0.427	0.446	0.437	0.396	0.351	0.303	0.289		
	taud		2.454	2.399	2.405	2.261	2.232	2.280	2.204	2.249	2.312	2.443	2.516	2.477		
	Ebn at Noon		278	287	290	279	272	270	264	263	266	269	272	270		
	Edh at Noon		25	30	34	42	44	42	45	42	37	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		591	887	1171	1518	1772	1985	1980	1741	1500	1006	678	506		
	RadStd		69	74	81	118	127	186	125	92	107	111	55	64		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page